Bosonisation of the Complex-boson realisation of W_∞

Source-backed arXiv sample

Abstract

We bosonise the complex-boson realisations of the W_∞ and $W_{1+\infty}$ algebras. We obtain nonlinear realisations of W_∞ and $W_{1+\infty}$ in terms of a pair of fermions and a real scalar. By further bosonising the fermions, we then obtain realisations of W_∞ in terms of two scalars. Keeping the most non-linear terms in the scalars only, we arrive at two-scalar realisations of classical w_∞.

1 Source Metadata

This sample is based on arXiv record `hep-th/9109004`. Authors: X. Shen and X.J. Wang.

2 Extracted Prose 1

0= 0= 0= 1= 1= 1= 2= 2= 2= 3= 3= 3= = = = = = = = = =.5em plus .25em minus .15em =14pt =14pt plus 3pt minus 10pt =14pt plus 3pt minus 10pt =0pt plus 3pt =8pt plus 3pt minus 5pt =3pt plus 1.5pt = = = #1

3 Extracted Prose 2

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